Sai Kaushik .P

ai23btech11017@iith.ac.in | sai2006kaushik@gmail.com | +91 9346424572 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.20
Sri Surya Junior College	Class XII (BIEAP)	2023	97.50%
St. Vincent De Paul School	Class X (CISCE)	2021	97.60%

SKILLS

- **Programming Language**: Python, C, Cpp, Java, RISC-V assembly
- **Tools**: Git, LaTeX, Solid Edge, Linux

WORK EXPERIENCE

AGC Asia Pacific Pte Ltd | Data Scientist Internship

Project: Flow-Analytics

- Developed an ML-driven traffic-risk analytics pipeline for **near-miss detection and risk ranking**, combining rule-based safety indicators with feature-based anomaly/risk modeling.
- Built and evaluated **unsupervised and supervised ML workflows** for prioritizing high-risk interactions, with visualization and replay-based analysis to support downstream review.
- Optimized large-scale trajectory-data processing using **memory-efficient, targeted data extraction**, significantly improving analysis throughput and enabling automated spatial risk visualization.

PROJECTS

Adv. AI Techniques for Long Video Understanding

Under the guidance of Prof. Konda Reddy M and Sunayna Padhye, M.Tech AI'26

- Contributed to a team of 3 to conduct in-depth research on AI and machine learning methodologies, Designing and implementing scalable algorithms to efficiently process and analyse large-scale video data, focusing on improving system performance and robustness.

Robust Motion-Gated Adaptation

Video Understanding & Test-Time Adaptation

- **Improved 3D video model robustness by +8.80% average top-1 accuracy** (up to **73.22% on MC3-18**) by developing a test-time adaptation algorithm (RMGA) that selectively updates BatchNorm layers using temporal sliding-window peak anchoring.
- **Reduced prediction uncertainty on corrupted streams by 9.5% in mean prediction entropy** while gaining up to **+13.57% accuracy on dynamic action classes** by implementing real-time motion-gated spatial masking via -thresholded frame differencing.
- **Engineered a reproducible evaluation suite spanning 50 action classes (6,680+ clips)** across multiple 3D backbones by building PyTorch pipelines with multi-worker synthesis, Replay mapping, and automated statistical telemetry.

Multi-Spectral Satellite Image Segmentation

Deep Learning Course

- **Designed a dual-branch DNN** for 16-channel multispectral image segmentation; built custom residual blocks with channel attention to prevent gradient issues and strengthen feature fusion.
- **Implemented a weighted compound loss** with deep supervision, accelerating convergence and mitigating class imbalance- improving validation IoU and Dice on multispectral tiles.

- **Developed a memory-efficient pipeline** streaming gigabyte-scale imagery via zero-copy memory-mapped loading, with threshold-calibrated inference for optimized performance.

Deep Learning-Based Coin Classification

Machine Learning Club Recruitment Project

- Built a multi-class coin image classifier using transfer learning in PyTorch, improving accuracy from **39.5% to 69.0%** on a dataset of 10,300+ coin images.
- Developed a robust PyTorch data pipeline with image augmentation and automated **handling of corrupted or unsupported files**, enabling reliable training across diverse image inputs.
- Evaluated model performance on 30 self-collected coin photographs under varied lighting, color, and camera angles to measure real-world generalization and identify failure cases.

RISC-V Simulator

Computer Architecture Course, Group Project

- **Developed an interactive 64-bit RISC-V instruction simulator in C**, enabling step-by-step program execution, breakpoint debugging, and call stack visualization by implementing assembly parsing, register tracking, and memory mapping.
- **Built a modular cache simulation framework** to analyze memory hierarchy performance and eviction trade-offs, supporting direct, set, and fully associative mappings alongside dynamic replacement policies (LRU, LFU, FIFO, and Random).
- **Integrated performance telemetry and profiling tools**, enabling quantitative evaluation of cache hit/miss rates and memory access patterns by implementing real-time cache state diagnostics and automated report generation.

RELEVANT COURSES

- **Artificial Intelligence**: Multi-Agent-Control-Systems, Computer Vision, Natural Language Processing, Data Mining, Deep Learning, Reinforcement Learning, Online Learning Algorithms, Robotics, Convex Optimization, Probability & Random Variables.

- **Computer Science** : Distributed AI Training, Software Engineering, Data Structures & Algorithms, Database Management Systems, Discrete Mathematics,Computer Architecture, Intro to Compilers, Intro to Operating Systems.